THE FUNCTOR AXIOMS IMPLY F IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2640-2651 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2653-2681, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_artin.json (pb=1544, stmt 2640-2651, proof 2653-2681).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 07WF: sources/stacks-artin/pinned-original-source.tex, source lines 2681–2681. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07WI: sources/stacks-artin/pinned-original-source.tex, source lines 2681–2681. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07XY: sources/stacks-artin/pinned-original-source.tex, source lines 2408–2421. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 02J4: sources/stacks-artin/pinned-original-source.tex, source lines 2681–2681. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07XZ: sources/stacks-artin/pinned-original-source.tex, source lines 2501–2619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 000R: sources/stacks-artin/pinned-original-source.tex, source lines 2681–2681. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let S be a locally Noetherian scheme. Let $F : (Sch/S)^{opp}_{fppf} \to Sets$ be a functor. Assume that*

1. *$\Delta : F \to F \times F$ is representable by algebraic spaces,*
2. *F satisfies axioms [-1], [0], [1], [2], [3], [4], [5] (see Section 07XZ), and*
3. *$\mathcal{O}_{S,s}$ is a G-ring for all finite type points s of S.*

Then F is an algebraic space.

Proof. Lemma 07XY applies to F. Using this we choose, for every finite type field k over S and $x_0 \in F(\mathrm{Spec}(k))$, an affine scheme U_{k,x_0} of finite type over S and a smooth morphism $U_{k,x_0} \to F$ such that there exists a finite type point $u_{k,x_0} \in U_{k,x_0}$ with residue field k such that x_0 is the image of u_{k,x_0}. Then

$$U = \coprod\nolimits_{k,x_0} U_{k,x_0} \longrightarrow F$$

is smooth[1]. To finish the proof it suffices to show this map is surjective, see Bootstrap, Lemma 07WF (this is where we use axiom [0]). By Criteria for Representability, Lemma 07WI it suffices to show that $U \times_F V \to V$ is surjective for those $V \to F$ where V is an affine scheme locally of finite presentation over S. Since $U \times_F V \to V$ is smooth the image is open. Hence it suffices to show that the image of $U \times_F V \to V$ contains all finite type points of V, see Morphisms, Lemma 02J4. Let $v_0 \in V$ be a finite type point. Then $k = \kappa(v_0)$ is a finite type field over S. Denote x_0 the composition $\mathrm{Spec}(k) \xrightarrow{v_0} V \to F$. Then $(u_{k,x_0}, v_0) : \mathrm{Spec}(k) \to U \times_F V$ is a point mapping to v_0 and we win. □

[1]Set theoretical remark: This coproduct is (isomorphic) to an object of $(Sch/S)_{fppf}$ as we have a bound on the index set by axiom [-1], see Sets, Lemma 000R.